THE COKERNEL OF A MAP FROM AN ADEQUATE FUNCTOR INTO A MODULE-VALUED FUNCTOR IS ADEQUATE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Every adequate functor is a quotient of a direct sum of linearly adequate functors. Same chapter, Lemma lemma-adequate-surjection-from-linear, invoked at proof lines 655-657. Reduces to the case F a direct sum of linearly adequate functors.
- A map from a linearly adequate functor into a module-valued functor lifts along a presentation. Same chapter, Lemma lemma-lift-map, invoked at proof lines 660-661. Supplies the maps A^{n_i} -> M used in the diagram.

Source reference. Original lemma 06UX: sources/stacks-adequate/pinned-original-source.tex, source lines 468–468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V0: sources/stacks-adequate/pinned-original-source.tex, source lines 605–605. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let $\varphi : F \to \underline{M}$ be a map of module-valued functors on Alg_A with F adequate. Then* $\mathrm{Coker}(\varphi)$ *is adequate.*

Proof. By Lemma 06UX we may assume that $F = \bigoplus L_i$ is a direct sum of linearly adequate functors. Choose exact sequences $0 \to L_i \to \underline{A^{\oplus n_i}} \to \underline{A^{\oplus m_i}}$. For each i choose a map $A^{\oplus n_i} \to M$ as in Lemma 06V0. Consider the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \bigoplus L_i & \longrightarrow & \bigoplus \underline{A^{\oplus n_i}} & \longrightarrow & \bigoplus \underline{A^{\oplus m_i}} \\ & & \downarrow & \swarrow & & & \\ & & \underline{M} & & & & \end{array}$$

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Consider the A-modules

$$Q = \mathrm{Coker}(\bigoplus A^{\oplus n_i} \to M \oplus \bigoplus A^{\oplus m_i}) \quad \text{and} \quad P = \mathrm{Coker}(\bigoplus A^{\oplus n_i} \to \bigoplus A^{\oplus m_i}).$$

Then we see that $\mathrm{Coker}(\varphi)$ is isomorphic to the kernel of $\underline{Q} \to \underline{P}$. $\square$